Predictive Modeling for Diabetes Progression Risk

Binary Classification with Explainability, Fairness Auditing & Drift Monitoring

Dataset: Pima Indians Diabetes Database (UCI ML Repository)

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1. Executive Summary

Type 2 diabetes is a progressive condition. Without early identification, patients accumulate complications — neuropathy, cardiovascular disease, kidney failure — that place significant strain on individuals and healthcare systems. The clinical value of an accurate risk model lies in enabling targeted intervention before complications develop.

This project builds a precision-focused binary classifier on the Pima Indians Diabetes Database that outputs a continuous risk score (predicted probability) as a proxy for progression risk. Fairness auditing, SHAP-based explainability, and drift monitoring are treated as first-class requirements. The model is served via a Streamlit clinician interface.

The best-performing model is a Support Vector Machine (SVM) with an RBF kernel, achieving a Test F0.5 of 0.711 and ROC-AUC of 0.839 on held-out data. The optimised decision threshold is 0.374.

F0.5 was chosen as the primary metric because false positives carry real clinical cost in an intervention-targeting setting. When the model flags a patient as high-risk, the downstream action must be justified. Precision is therefore weighted above recall.

2. ML Problem Formulation

The Pima Indians Diabetes Database provides a binary outcome variable (diabetes onset: yes or no). There is no continuous progression score to regress on, which rules out regression as a direct approach. Instead, the predicted probability from a binary classifier serves as a meaningful continuous progression proxy — a risk score in [0.0, 1.0] that quantifies each patient's likelihood of diabetes onset and acts as a ranked risk indicator for clinical triage.

F0.5 is chosen over F1 or F2 because this model targets intervention, not screening. Interventions carry cost and patient burden. A false positive triggers unnecessary follow-up and erodes clinician trust. F0.5 encodes the priority that when the model says 'high risk', it should be right.

Attribute	Decision
Task Type	Binary Classification
Target Variable	Outcome (0 = no onset, 1 = onset)
Progression Proxy	Predicted probability — Risk Score in [0.0, 1.0]
Primary Metric	F0.5 Score (precision-weighted)
Deployment	Streamlit clinician interface

3. Dataset & EDA Highlights

The dataset contains 768 records from female patients of Pima Indian heritage, each described by 8 clinical features and a binary target. The class distribution is moderately imbalanced at 35% positive (diabetic) and 65% negative — requiring explicit handling to prevent the model from defaulting to the majority class.

Missing Values

Five features contain biologically implausible zero values. Glucose, Blood Pressure, Skin Thickness, Insulin, and BMI cannot be zero in a living patient — these were treated as missing. Insulin had approximately 49% missing after this correction and Skin Thickness around 30%, making them the most challenging features to impute.

Key Discriminative Features

Glucose showed the strongest separation between diabetic and non-diabetic groups across all visualisations. BMI and Age were also highly informative. The Insulin-SkinThickness correlation of 0.66, visible in the heatmap below, directly motivated the choice of KNN imputation over median.

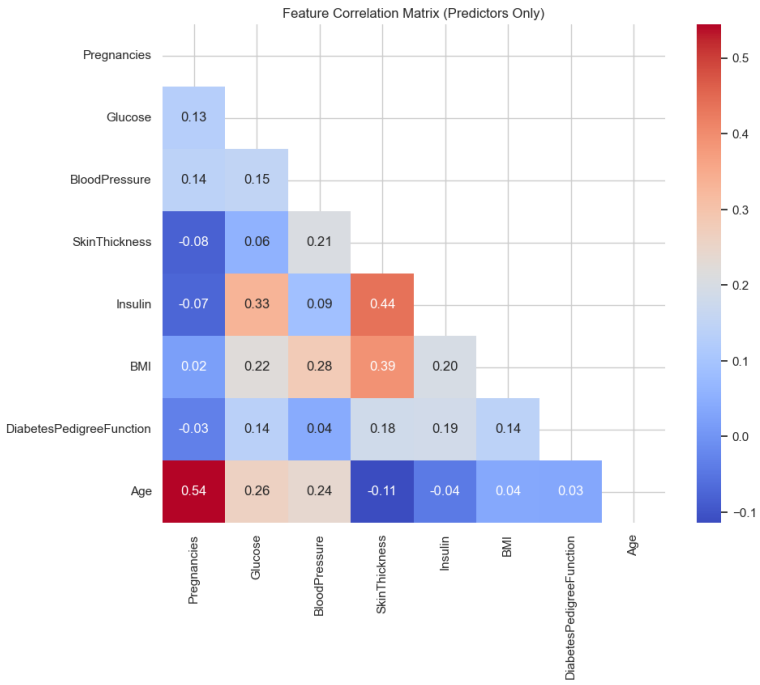

Figure 1: Feature correlation matrix. Insulin-SkinThickness correlation (0.44) and Age-Pregnancies (0.54) are the strongest pairs.

4. Preprocessing & Feature Engineering

The dataset was split 70/15/15 into train, validation, and test sets using stratified sampling before any imputation or scaling — the single most important leakage prevention step. A KNN Imputer was fitted on the training set only, exploiting the Insulin-SkinThickness correlation to reconstruct

missing values more faithfully than a univariate statistic.

Feature	Formula	Rationale
glucose_bmi_interaction	Glucose x BMI	Combined metabolic burden signal
insulin_glucose_ratio	Insulin / (Glucose + e)	Proxy for insulin resistance
age_group	Binned age buckets	Grouping for fairness audit
bmi_group	WHO BMI categories	Grouping for fairness audit

SMOTE was applied inside each CV fold via an imblearn Pipeline — never to the validation or test sets. This preserves the natural 35/65 imbalance in evaluation splits so that reported metrics reflect real-world distribution. StandardScaler was fitted on the training set only and applied to all splits.

5. Modelling & Selection

Five model families were evaluated: Logistic Regression (linear baseline), Random Forest (bagging ensemble), XGBoost (boosting ensemble), SVM (kernel method), and KNN (instance-based). RandomizedSearchCV (50 iterations) was followed by a narrowed GridSearchCV, all scored on F0.5 with 5-fold stratified CV.

Model	Val F0.5	Train F0.5	Gap	Val Loss	Decision
LR Baseline	0.6048	0.6590	0.054	0.558	Honest floor
Random Forest	0.5943	0.8763	0.282	0.519	Discarded — overfit
XGBoost	0.6042	0.7498	0.146	0.531	Discarded — see below
SVM (RBF)	0.6360	0.7393	0.103	0.515	SELECTED
KNN	0.6250	0.7043	0.079	2.270	Discarded — miscalibrated

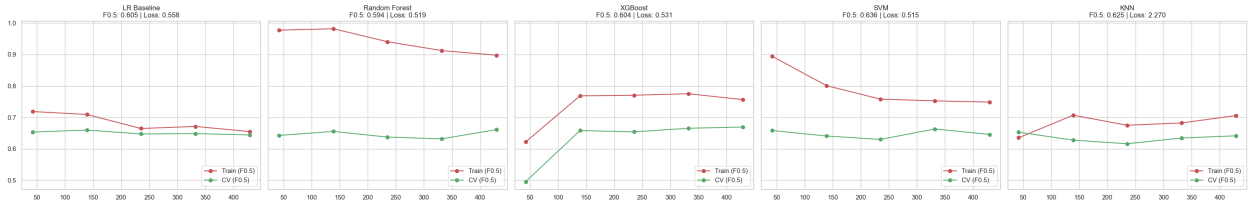

Figure 2: Learning curves for all five models. SVM (4th panel) shows the cleanest convergence. Random Forest and XGBoost show persistent train/CV gaps indicating overfitting.

Why SVM Over XGBoost

XGBoost's first pass had a train/val gap of 0.340 — severe memorisation. A second search extended to regularisation parameters (subsample, colsample_bytree, min_child_weight, reg_alpha, reg_lambda). The gap closed to 0.146 but validation F0.5 fell from 0.636 to 0.604. With 768 rows and 10 features, XGBoost's boosting capacity finds no additional signal to reward — it constrains itself without improving. SVM achieves the same validation F0.5 as XGBoost's peak with a substantially cleaner bias-variance profile and stable convergence.

Why KNN Was Discarded

KNN's validation log-loss of 2.270 signals severely miscalibrated probability outputs. Since the system returns a risk score rather than only a binary label, unreliable probabilities disqualify it entirely.

Threshold Optimisation

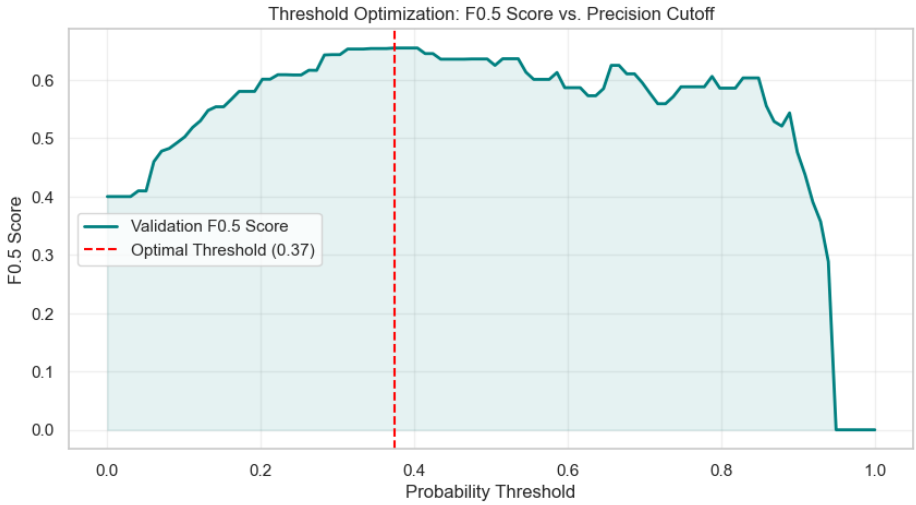

Figure 3: F0.5 score across candidate thresholds on the validation set. Optimal threshold: 0.374 (dashed red line).

The optimal threshold of 0.374 is lower than the conventional 0.5 — indicating the SVM's probability outputs skew below 0.5 for this dataset, and the optimal operating point sits to the left of centre. Clinically, this threshold was selected to maximise F0.5 on validation data and is applied consistently at inference time.

6. Evaluation Results

The test set was evaluated once, after all model selection and threshold decisions were finalised on the validation set.

Metric	Score	Interpretation
F0.5 Score	0.711	Primary metric; precision-weighted performance on held-out data.

Metric	Score	Interpretation
ROC-AUC	0.839	Strong discriminative ability across all thresholds.
PR-AUC	0.726	Robust positive-class performance under imbalance.
Precision	0.696	Of patients flagged high-risk, ~70% are true positives.
Recall	0.780	Of all true diabetics, ~78% are correctly identified.

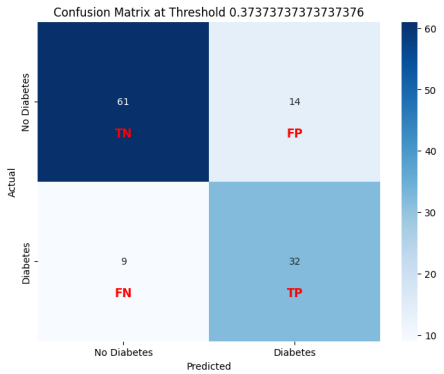

Figure 4: Confusion matrix at threshold 0.374. 61 TN, 14 FP, 9 FN, 32 TP.

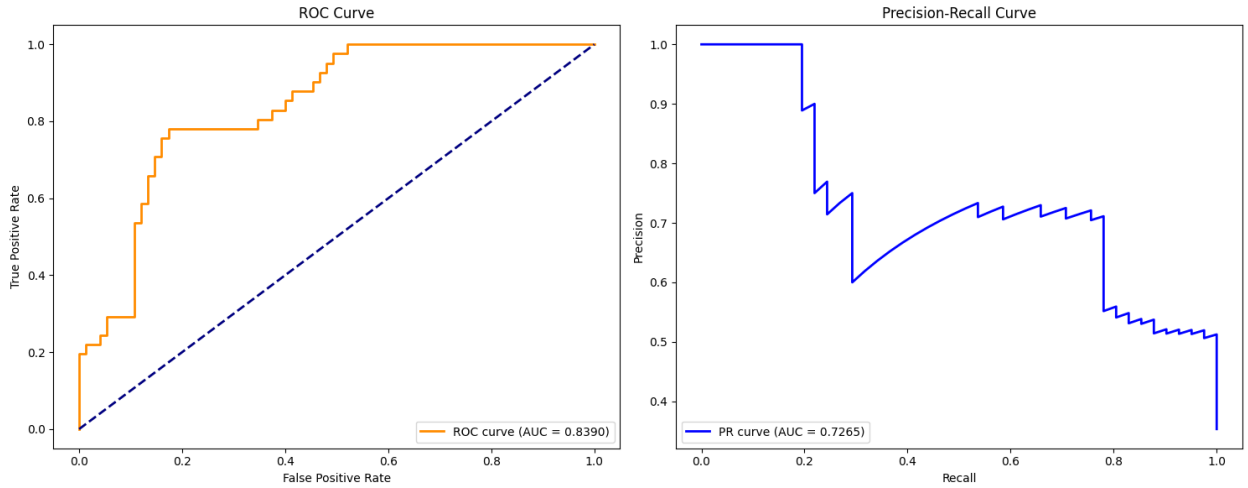

Figure 5: ROC curve (AUC 0.839) and Precision-Recall curve (AUC 0.726). PR-AUC is the primary curve for imbalanced datasets as it excludes true negatives from its computation.

Calibration Decision

A CalibratedClassifierCV (sigmoid, 5-fold) was applied and evaluated. Recalibration improved probability linearity but reduced Test F0.5 from 0.711 to 0.567 — a drop of 14.4 percentage points. The uncalibrated model was retained. Clinicians should interpret risk scores as relative rankings rather than exact probability estimates.

7. Fairness Audit

Fairness was evaluated across four sensitive attribute groups: age_group, bmi_group, pregnancy_group, and glucose_group. Gender is unavailable as all patients in the dataset are female. Alert thresholds: Equalized Odds Difference > 0.10 and Precision Parity gap > 0.10.

Age Group	Precision	Recall	F0.5	FPR	FNR
20-30	0.643	0.529	0.616	0.094	0.471
31-45	0.737	1.000	0.778	0.455	0.000
46-60	0.692	0.900	0.726	0.500	0.100
60+	0.000	0.000	0.000	0.000	0.000

The 60+ group shows zero across all metrics — a sample size artifact with only 3 patients in that bracket in the test set. Performance across the 20-60 range is stable. The aggregate Equalized Odds Difference of 1.000 for age_group is driven entirely by this small-sample edge case and does not reflect a systematic failure for older patients.

BMI Group	Precision	Recall	F0.5	FPR	FNR
Normal	0.500	0.500	0.500	0.048	0.500
Overweight	0.500	0.500	0.500	0.130	0.500
Obese	0.737	0.848	0.757	0.323	0.152

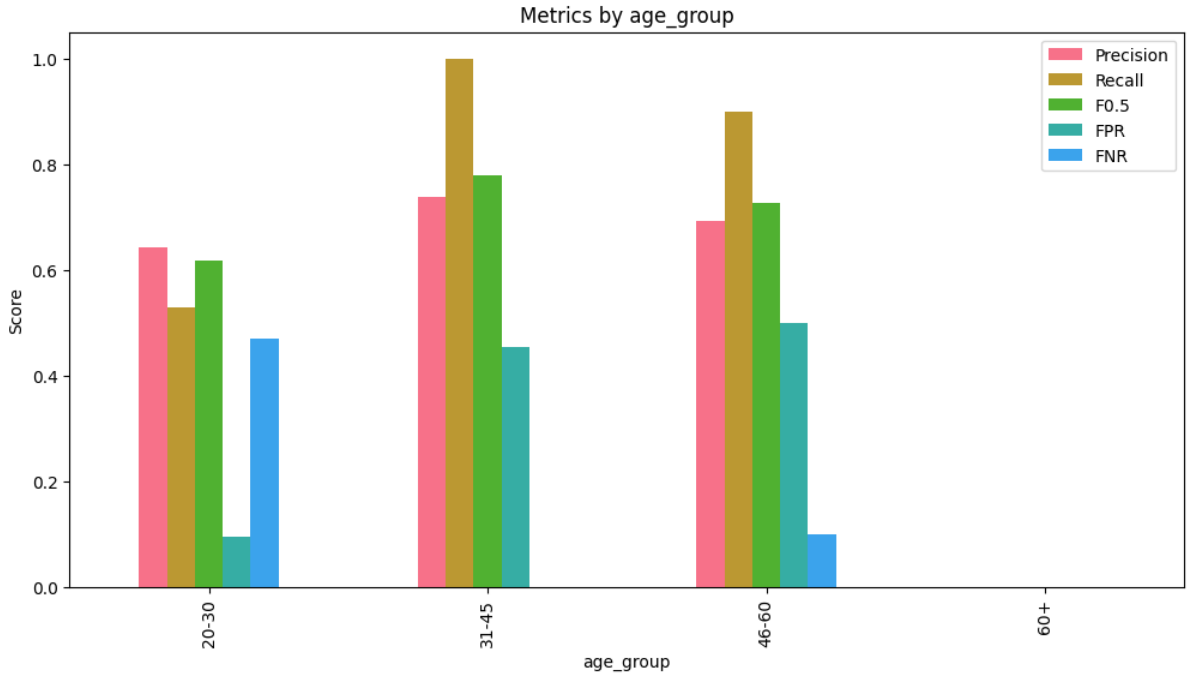

Figure 6: Per-metric performance by age group. The 60+ bar is empty due to n=3 in the test set.

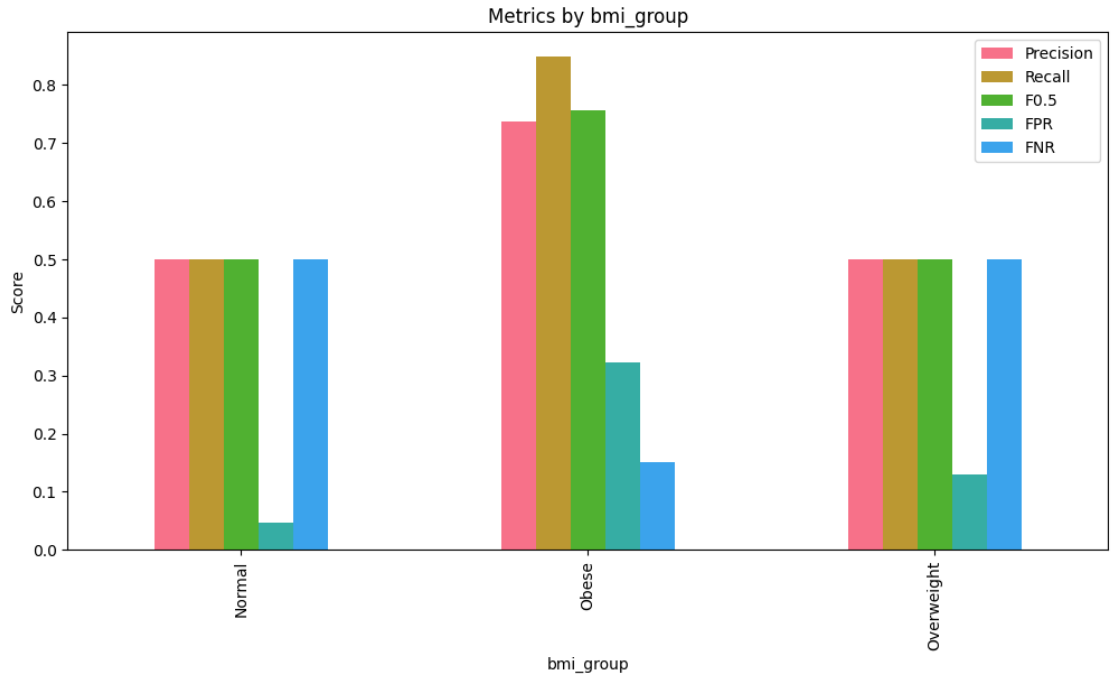

Figure 7: Per-metric performance by BMI group. The Obese group shows materially stronger performance, reflecting higher diabetic prevalence in that bracket.

Attribute	Dem. Parity Diff	Eq. Odds Diff	Precision Gap	Status
age_group	0.778	1.000	0.737	60+ artifact (n=3)

Attribute	Dem. Parity Diff	Eq. Odds Diff	Precision Gap	Status
bmi_group	0.257	0.349	0.237	Real disparity — document

The BMI disparity is a genuine finding. The model performs materially better for Obese patients than for Normal or Overweight patients. In deployment, clinicians should be made aware that confidence is lower for patients in the Normal BMI range — a positive flag for such patients warrants closer human review. No algorithmic mitigation was applied: that decision belongs with clinical stakeholders who can weigh the fairness-accuracy tradeoff.

8. Explainability

SHAP KernelExplainer was used throughout. SVM models are not tree-based and are therefore incompatible with TreeExplainer. KernelExplainer is model-agnostic, approximating Shapley values against a background dataset of 50 randomly sampled training observations.

Global Feature Importance

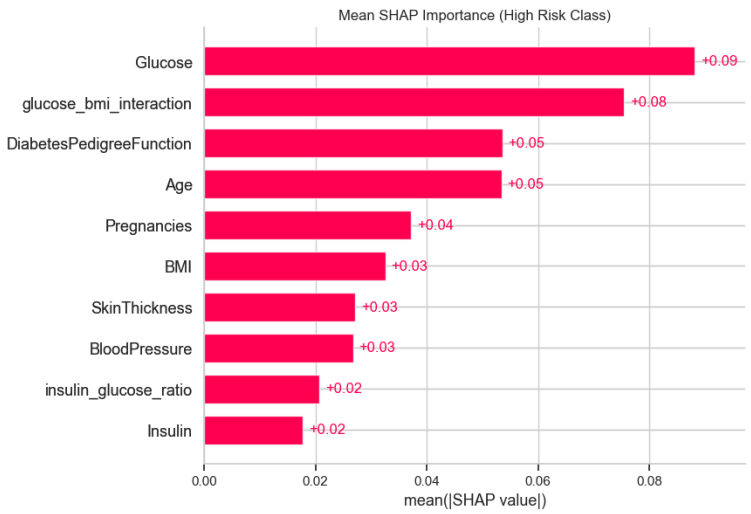

Figure 8: Mean absolute SHAP values by feature. Glucose dominates, followed closely by glucose_bmi_interaction — confirming the engineered feature adds signal beyond either raw feature alone.

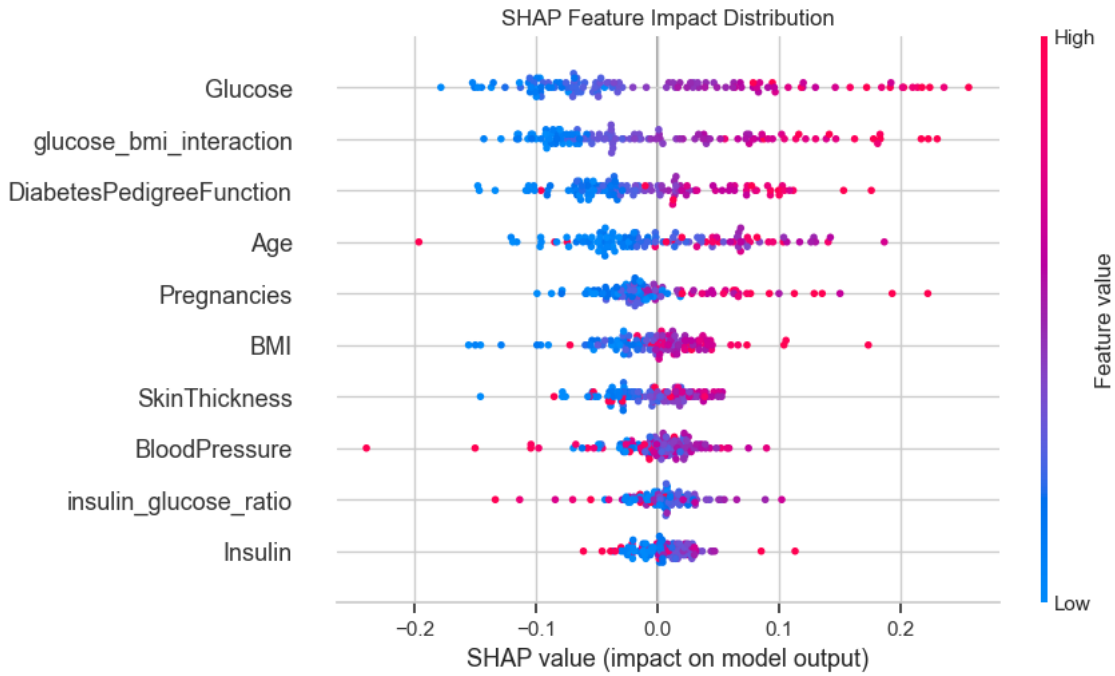

Figure 9: SHAP beeswarm plot. Each point is one test patient. Red = high feature value, blue = low. High Glucose consistently pushes predictions toward high risk (positive SHAP).

The top three features by mean absolute SHAP are Glucose, glucose_bmi_interaction, and Diabetes Pedigree Function — consistent with clinical literature. The engineered interaction feature ranked second globally, confirming that combined metabolic burden carries additional predictive signal beyond each feature individually.

Local Explanation — High-Risk Patient

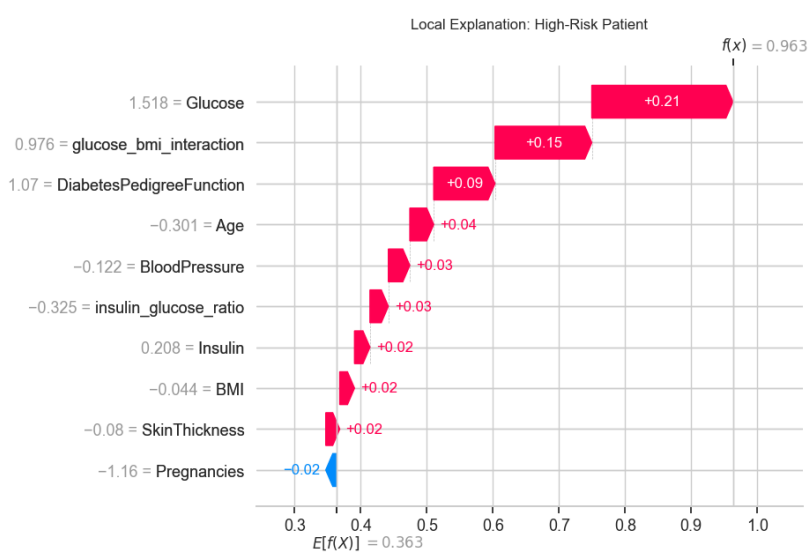

Figure 10: SHAP waterfall for a high-risk patient (predicted probability: 0.963). Glucose (+0.21) and glucose_bmi_interaction (+0.15) are the dominant contributors. Pregnancies is the only protective factor (-0.02).

This waterfall format is reproduced in the Streamlit interface as a natural-language summary with a SHAP bar chart, allowing clinicians to understand the specific drivers behind each individual prediction without requiring statistical literacy.

9. Drift Monitoring

Drift monitoring uses Population Stability Index (PSI) as the primary signal and the Kolmogorov-Smirnov test as a secondary check. The baseline distribution is the scaled training set (537 patients).

PSI Range	Status	Action
< 0.10	Stable	No action required
0.10-0.20	Warning	Investigate; increase monitoring frequency
> 0.20	Alert	Model review recommended before continued use

Real Data Baseline Check (X_train vs X_test)

Insulin showed a PSI of 0.253 and Diabetes Pedigree Function 0.553 between the training and test splits drawn from the same dataset. Values this high from the same source data reflect sensitivity of PSI to small sample sizes (~115 test observations). In production with larger volumes, PSI estimates would stabilise — and alert thresholds may need recalibration for this dataset scale.

Drift Simulation

- Scenario 1 — Glucose +20 mean shift: Triggered alerts on Glucose and glucose_bmi_interaction. The alerting system correctly propagates upstream feature drift to downstream engineered features.
- Scenario 2 — BMI +3 and Age +5 shift: Triggered alerts on BMI, Age, glucose_bmi_interaction, and insulin_glucose_ratio. This scenario better reflects realistic population ageing or cohort composition change.

Recommended Deployment Schedule

Weekly PSI checks against the stored baseline are recommended. Any feature exceeding PSI 0.20 should trigger a model review before the next clinical decision cycle. Drift alerts are surfaced to model owners, not clinicians.

10. Deployment Architecture

The model is deployed as an interactive clinical frontend built with Streamlit. This eliminates a separate backend layer, reducing deployment friction and making the system immediately usable for evaluation.

Data Ingestion and Processing

A clinical sidebar accepts the 8 core physiological parameters. On submission, feature engineering (Glucose-BMI interaction, Insulin-Glucose ratio) and StandardScaler normalisation are applied in-memory using the saved preprocessing artifacts, exactly replicating the training pipeline.

Dashboard Outputs

- Risk Score and Classification: The model outputs a continuous risk probability and classifies each patient as HIGH RISK or LOW RISK against the optimised threshold of 0.374.
- Dynamic SHAP Explanation: A live SHAP KernelExplainer generates a patient-level horizontal bar chart quantifying which biomarkers pushed the prediction toward or away from the baseline risk. Explanations are computed per submission, not precomputed.
- Natural Language Summary: SHAP values are parsed into a readable narrative identifying primary risk drivers and protective factors, formatted for rapid clinician consumption without requiring statistical literacy.
- Automated Reliability Advisories: Runtime warnings are triggered when a patient falls into a sparsely-represented cohort — specifically: Age > 60, Glucose < 100, or BMI < 25.0. This operationalises the fairness audit findings at the point of clinical use.

Production Path

Streamlit is appropriate for prototyping and internal tooling. A production deployment at scale would separate inference logic into a dedicated API layer with Streamlit or a purpose-built frontend consuming it. The current architecture is designed with that separation in mind: preprocessing, prediction, and explanation logic are encapsulated in standalone modules that can be wrapped by an API endpoint without refactoring.

11. Ethics & Limitations

Single-gender, single-ethnicity dataset

All 768 records are from female patients of Pima Indian heritage. The model cannot be generalised to male patients or other ethnic populations without retraining on representative data. Any deployment must be scoped explicitly to populations represented in the training data.

No temporal data

The dataset is cross-sectional — each record is a single snapshot. Diabetes progression is a longitudinal process. The progression proxy in this project approximates onset likelihood, not rate

of change over time.

Simulated drift validation

Drift monitoring was validated against synthetic distribution shifts, not real population change. Production deployment requires live data logging and longitudinal PSI tracking to assess real drift behaviour.

Diabetes Pedigree Function opacity

The formula encoding family history into the DPF score is not fully published. Its weighting of different family relationships cannot be audited for bias, limiting the completeness of the fairness analysis.

Fairness attribute coverage

Only age and BMI groups were available as sensitive attributes. Socioeconomic status, rural vs. urban access to care, and ethnicity beyond Pima heritage are absent from the dataset and cannot be audited.

Probability calibration

The uncalibrated SVM was retained for its superior F0.5 performance. Risk scores should be interpreted as ordinal rankings rather than precise probability estimates.

12. Conclusion

This project delivers a complete end-to-end machine learning pipeline for diabetes progression risk prediction: from exploratory analysis and preprocessing through model selection, evaluation, fairness auditing, explainability, drift monitoring, and a deployed Streamlit clinician interface.

The selected model — SVM with RBF kernel — achieves a Test F0.5 of 0.711 and ROC-AUC of 0.839, with an optimised decision threshold of 0.374. Its selection over XGBoost reflects a deliberate, evidenced conclusion: on small structured datasets, a well-tuned kernel method with a clean bias-variance profile outperforms high-capacity ensembles that lack sufficient data to justify their complexity.

Fairness and explainability were integrated as core requirements. The fairness audit identified a genuine performance disparity across BMI groups, documented transparently with a recommendation for clinician awareness. SHAP explanations provide patient-level interpretability surfaced directly in the clinical interface, including automated runtime warnings for under-represented patient cohorts.

The primary path for future improvement is longitudinal data. With time-series records per patient, this framework could extend to true progression modelling — predicting rate of change rather than onset likelihood — and the deployment architecture is designed to accommodate that extension.